

# Firewall Basics

## Task 1: Block Spotify Using Windows Defender Firewall

I used Windows Defender Firewall to create an outbound rule to block traffic from the Spotify desktop application.

### Steps:

1. Open **Control Panel**.
2. Go to **System and Security** → **Windows Defender Firewall**.
3. Click **Advanced settings**.
4. Select **Outbound Rules**.
5. Click **New Rule**.
6. Select **Program** and click Next.
7. Select **This program path** and browse to the Spotify application file.
8. Select **Block the connection**.
9. Keep the required network profiles selected.
10. Give the rule a name such as **Block Spotify**.
11. Click **Finish**.

After creating the rule, Windows Firewall will block outbound traffic from the selected Spotify application.

## **Allow Google Chrome on Port 443**

I created an inbound firewall rule to allow HTTPS traffic on port 443 for Google Chrome.

### **Steps:**

1. Open **Windows Defender Firewall with Advanced Security**.
2. Click **Inbound Rules**.
3. Select **New Rule**.
4. Select **Port** and click Next.
5. Select **TCP**.
6. Select **Specific local ports** and enter **443**.
7. Select **Allow the connection**.
8. Select the required network profiles.
9. Give the rule a name such as **Allow Chrome HTTPS 443**.
10. Click **Finish**.

Port 443 is commonly used for secure HTTPS communication

## Block a Specific IP Address

To block all traffic from the IP address **123.45.67.89**, I can create a custom inbound or outbound firewall rule.

### Steps:

1. Open **Windows Defender Firewall with Advanced Security**.
2. Select **Inbound Rules** or **Outbound Rules** depending on the traffic that needs to be blocked.
3. Click **New Rule**.
4. Select **Custom** and click Next.
5. Keep the default program option and continue.
6. In the **Scope** section, find **Remote IP address**.
7. Select **These IP addresses**.
8. Click **Add** and enter **123.45.67.89**.
9. Click Next.
10. Select **Block the connection**.
11. Select the required network profiles.
12. Give the rule a name such as **Block Specific IP**.
13. Click **Finish**.

The firewall rule will block the selected traffic to or from the specified IP address, depending on whether the rule is inbound or outbound.

## Hardware Firewall vs Software Firewall

| Point         | Hardware Firewall                                                                   | Software Firewall                                                                    |
|---------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1. Location   | It is a separate physical device placed between the network and the internet.       | It is installed and runs on a computer or operating system.                          |
| 2. Protection | It can protect multiple devices on the network at the same time.                    | It mainly controls traffic for the individual device where it is installed.          |
| 3. Management | It is usually managed by a network administrator and is useful for larger networks. | It can usually be configured directly on the computer by the user or administrator.  |
| Use Case      | Used in offices, schools and large organizations to protect the complete network.   | Used on personal computers and laptops to control incoming and outgoing connections. |

## **Conclusion**

A firewall helps protect a computer or network by controlling network traffic. Hardware firewalls are useful for protecting many devices in a network, while software firewalls provide protection directly on individual computers.